FLEET TEAM DIGEST

737MAX-FTD-32-17029

Issue Title : Wheel Bearing Failure Due to Improper Lock Washer Installation

Airplane Model	ATA	Minor Model(s)
737MAX	3240-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
12/04/2017	12/04/2017	04/21/2017

Estimated Completion Date	Next Update Date

Applicability

All 737MAX airplanes.

Description

Several 737NG operators have reported cases where the main landing gear (MLG) axle nut lock washer slipped out of position during tightening of the axle nut, resulting in a loose MLG wheel installation. A loose wheel installation can result in a wheel bearing failure.

Background

During a MLG wheel change on a 737-800, an operator reported that the axle nut was tightened to the required torque value and the wheel was found to be loose. Upon removal of the wheel, the mechanic found that the axle nut lock washer had rotated, disengaging the washer tang from the keyway groove in the axle.

A review of the installation found that the washer can slip out of the keyway groove and rotate if the axle nut is used to run the wheel and washer onto the axle. When the washer tang reaches the thread relief on the axle, the washer may rotate with the nut as the nut is tightened. The tang can then hang up on the edge of the axle journal outside the keyway groove, giving a false indication that the wheel is properly installed.

Status

Closed

Interim Action

N/A

Final Action

Boeing recommends that both the wheel and the lock washer are firmly pushed all the way onto the axle until the wheel is seated prior to installing and tightening the axle nut. Ensure that the lock tang is fully engaged in the keyway groove in the axle. Avoid using the axle nut to push the wheel and lock washer into place.

In addition, AMM 32-45-11, step 3.H.(1)(b) instructs to apply a thin layer of grease to the faces of the washer, prior to being

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installed on the axle. This is to help hold it in place against the wheel bearing until the nut is tightened.

AMM Chapter 32-45-11, Task 32-45-11-400-801, Main Landing Gear Wheel and Tire Assembly Installation, includes the above information in the wheel installation instructions.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Aircraft Maintenance Manual (AMM)	32-45-11		

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- In Service Occurrences : null

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